

Don Norman's Work on Design

Books (Authored & Edited)

- **Design for a Better World (2023)** – Norman's latest book advocating "*humanity-centered*" design, focusing on meaning, sustainability, and designing for **all** of humanity (not just users) to address global challenges ¹ ² .
- **The Design of Everyday Things (1988; Revised 2013)** – Norman's seminal bestseller on user-centered design. It illustrates how poor design (e.g. confusing doors or controls) frustrates users and outlines principles (visibility, feedback, natural mappings, constraints) for "*good, usable design*" ³ . (Originally titled *The Psychology of Everyday Things*).
- **Emotional Design: Why We Love (or Hate) Everyday Things (2004)** – Explores the emotional side of design. Norman shows through research that *attractive things work better*, articulating how products' visceral, behavioral, and reflective qualities influence our feelings and behavior ⁴ .
- **Living with Complexity (2010)** – A book debunking the myth that simplicity is always the goal. Norman argues *complexity isn't the enemy – bad design is*. Good design can tame complexity to make systems understandable and manageable, matching the inherent complexity of real life ⁵ .
- **The Design of Future Things (2007)** – Examines "smart" technologies and the future of human-machine interaction. Norman discusses talking cars, intelligent appliances, and the pitfalls of automation, proposing design principles for natural, consumer-friendly interaction with autonomous systems ⁶ .
- **The Invisible Computer (1998)** – An analysis of why personal computers were too complex for many users. Norman advocates a shift to "**information appliances**" – simpler, task-focused devices – and a human-centered product development process ("*user needs first, technology last*") even if it means reorganizing companies ⁷ .
- **Things That Make Us Smart (1994)** – Investigates how technology and cognition intersect. Norman argues for machines that *fit our minds* instead of forcing people to conform to machines, urging a person-centered redesign of the tools we use ⁸ . It introduced ideas like **cognitive artifacts** and how design can support human memory and problem-solving.
- **Turn Signals Are the Facial Expressions of Automobiles (1993)** – A collection of essays on technology's foibles. Norman uses everyday examples (from **water faucets** to **airplane cockpits**) to show how many gadgets seem designed with little regard for humans. A sometimes whimsical, often pointed call for more "*humanization of modern design*" ⁹ .
- **User Centered System Design (1986)** – An edited volume (Norman & Draper) that introduced the term "*user-centered design*." This academic book compiled new perspectives on human-computer interaction, arguing that designers must **focus on users' needs and tasks** at every step ¹⁰ ¹¹ . (*It was a "significant milestone" in HCI, helping shift design thinking toward the user* ¹² .)
- **Early Academic Works (1960s–70s)** – Before focusing on design, Don Norman was a cognitive psychologist. He authored foundational texts like *Human Information Processing: An Introduction to Psychology* (1972), *Memory and Attention* (1st ed. 1969), *Explorations in Cognition* (1975, with D. E. Rumelhart), and *Learning and Memory* (1982) ¹³ ¹⁴ . These works, while technical, informed his later design theories by deepening understanding of human memory, learning, and mental models.

Notable Articles & Essays

- **“The Truth About Unix: The User Interface is Horrid” (1981)** – A provocative early article in *Datamation* where Norman critiqued the usability of Unix systems. This piece (infamous for its title) catapulted him to prominence in HCI by bluntly showing how tech-centric design neglects human users ¹⁵.
- **“Human-Centered Design Considered Harmful?” (Interactions, 2005)** – A controversial essay arguing that *traditional HCD is too narrow*. Norman suggested designers should shift from focusing only on individual users (and personas) to **activity-centered design** – looking at the larger contexts and workflows. In a later clarification, he explained that HCD had become “*a limited view...focused upon screen-by-screen analysis*”, missing broader sequences and error situations in real activities ¹⁶. This spurred extensive debate on balancing user needs with overall human **activities** and systems.
- **“Design Thinking: A Useful Myth” (Core77, 2010)** – An essay in which Norman challenged the hype around “design thinking.” He argued that designers do not have a mystical process exclusive to them – great scientists, engineers, and artists use similar creative problem-solving. The “*myth*” is that design thinking alone guarantees innovation ¹⁷. (Norman later softened this stance, calling design thinking an “*essential tool*” but cautioning against idealizing it ¹⁸.)
- **“Emotion & Design: Attractive Things Work Better” (2002)** – A short piece (preceding the *Emotional Design* book) where Norman first publicized his finding that positive emotion can enhance usability. It used examples like how a **pleasant aesthetic** can reduce stress and thus improve users’ cognitive performance. (Cited in *Scientific American* and elsewhere ⁴.)
- **Core77 Columns (2011–2013)** – Norman wrote an influential series of online columns for Core77. Topics included *design education* (“*Brilliance Without Substance*”), cultural differences in design (“*Does Culture Matter?*”), and critiques of tech trends (e.g. “*Gesture Wars*” on inconsistent touch gestures). In “*Rethinking Design Thinking*” he revisited his earlier essay, ultimately acknowledging design thinking’s value when properly applied ¹⁸. These essays disseminated his ideas to practicing designers and often sparked community discussion.

Talks & Lectures

- **TED Talk – “3 Ways Good Design Makes You Happy” (TED2003)** – In this popular 12-minute talk (posted 2009), Don Norman explores the relationship between design and emotion. He uses engaging examples (like an aesthetically pleasing tea kettle) to show how *beauty, fun, and pleasure* in design can actually make products more effective ¹⁹. (This talk distilled concepts from *Emotional Design* for a broad audience.)
- **Keynotes & Conference Talks** – Norman is a sought-after keynote speaker. For example, at UX conferences he has spoken on topics like **21st-century design challenges**, the impact of AI, and the future of design education. In a 2023 McKinsey Author Talk (a video interview), the former Apple VP urged business leaders to adopt “*humanity-centered design*” and discussed collaborating with **generative AI** and building a circular economy ²⁰. At universities (e.g. Stanford, Harvard), he’s lectured on themes from his books such as *complexity* (Stanford seminar on *Living with Complexity*) and *future technology* (e.g. “**Design of Future Things**” talk at Google).
- **Design@Large and Other Series** – As founding director of UCSD’s Design Lab, Norman often engaged in public dialogues. In the UCSD *Design@Large* series, he conversed with other design leaders about designing for global change. Similarly, an ACM CHI conference “**oral history**” interview (2021) captured Norman reflecting on his career – from cognitive science to Apple to the present – offering insights to HCI researchers ²¹. (These recorded talks and interviews form an educational resource for new designers.)
- **Media Appearances** – Norman has also appeared in popular media to explain design to general audiences. For instance, Vox’s “**Doors & Norman Doors**” video featured him explaining why **confusing doors** (now nicknamed “Norman doors”) still plague buildings ²². Such appearances

have helped spread basic design literacy (e.g. the idea that if a door needs a “Push/Pull” sign, it’s a design failure). He’s also been on design podcasts and TV segments, always advocating for the user’s perspective.

Interviews

- **McKinsey “Author Talks” (2023)** – In a recent interview, Don Norman (at age 87) discusses *Design for a Better World*. He explains why we must expand from human-centered to **humanity-centered design**, considering global systems and sustainability. *“If we make products that destroy the planet, that’s not good,”* Norman quips, calling for designers to focus on long-term human and environmental well-being ² ²³. This interview also touches on emerging tech like generative AI and how Norman envisions designers collaborating with AI tools.
- **UX Magazine Interview (2014)** – In a Q&A titled *“Solving the Right Problem...”*, Norman emphasizes the importance of defining the correct problem before designing solutions. He recounts examples like VCRs: the real goal was *“I want to record a show,”* not to set a clock – which led to the invention of the DVR ²⁴ ²⁵. He advises designers to keep asking “Why” (the **Five Whys** technique) to uncover fundamental user needs. This interview also covers team dynamics and why copying competitors can be a trap – *“don’t just match features; play to your product’s unique strengths.”* ²⁶
- **ACM SIGCHI Oral History (2021)** – A career-spanning interview (part of ACM’s archival project) in which Norman shares personal stories – from his time at UCSD founding the Cognitive Science department to his stint as an Apple VP. He reflects on mentors, mistakes, and his evolving design philosophy. *(Referencing this offers readers a deep dive, though the full transcript is behind ACM’s paywall.)*
- **Other Noteworthy Interviews** – Norman’s insights appear in many publications. A 1995 *Interactions* magazine piece **“Ask Don”** captured his candid answers to designers’ questions (many still relevant decades later). In **Design Observer** and **Core77** chats, he mused about the future of design research and why *“designers must tackle big, messy problems.”* And in a 2019 **Fast Company** interview, he provocatively critiqued design education, calling many design schools *“embarrassingly obsolete”* – urging updates to curricula. Across these interviews, Norman’s frank, often humorous style comes through, as does his core message: **design is about improving lives**, and we must keep humans (indeed, humanity) at the center ²⁷.

Teaching & Educational Contributions

- **Academic Roles** – Don Norman has had a profound impact on design education. He was founding chair of UCSD’s Department of Cognitive Science and later a professor of Computer Science and Design at Northwestern University. At Northwestern, he co-directed the **Segal Design Institute**, integrating human-centered design into engineering curricula. (He’s now *emeritus* at both UCSD and Northwestern.)
- **UCSD Design Lab** – In 2014 Norman returned to UC San Diego to launch the **Design Lab**. As director, he championed interdisciplinary design education addressing real-world problems. The Lab’s courses and projects (from healthcare UX to sustainability) were often guided by Norman’s principles. *For example, in 2022 he taught a class using a draft of *Design for a Better World*, challenging students to apply humanity-centered design to climate change and social issues* ²⁸. The Lab continues to embody Norman’s vision of designers as agents of positive change.
- **Nielsen Norman Group** – In 1998, Norman co-founded the **Nielsen Norman Group (NN/g)** with Jakob Nielsen. Through NN/g, he helped shape UX professional education – creating research-based **seminars, reports, and certification courses** that have trained tens of thousands of UX practitioners. Many NN/g course topics (like *The Human Mind and Usability*, *Emotion & Design*) draw on Norman’s work. Even if he isn’t always the instructor, his influence pervades NN/g’s teaching materials ²⁹ ³⁰.

- **Online Courses & Masterclasses** – Recently, Norman has lent his expertise to online education. In 2023 he partnered with the Interaction Design Foundation for a course “**Design for a Better World**” (based on his book), where students learn to apply human-centered design to global challenges ³¹ ³² . He has appeared in MOOC videos (e.g. Coursera’s *Human-Centered Design* intro) and in Masterclasses on UX. Through these, a new generation of designers worldwide can effectively “*learn from Don.*”
- **Mentorship and Writings** – Norman’s educational impact also comes via his writings and mentorship. His classic books are often **required reading** in HCI and design programs globally. He’s known for generously advising students and young designers (often via email or his website’s “**Ask Don**” column). Many of his shorter essays (freely available on JND.org) are used as teaching tools – e.g. “*Making a thermostat that teaches sustainability*” or “*Design X: Complex Societal Problems*”, which push students to think beyond apps and websites. In summary, beyond his direct teaching, Don Norman’s *ideas* have become part of the educational canon for design.

AI Design and UX Resources

(This section compiles key resources – both foundational and cutting-edge – for AI-focused design and user experience. It spans books and papers, design guidelines, toolkits, courses, practical how-tos, and community influences relevant to designing human-centered AI and GenAI interfaces.)

Books & Academic Papers on AI + Human-Centered Design

- **Human-Centered AI** – Ben Shneiderman (2022). A leading HCI researcher provides an “*optimistic realist’s*” guide on designing AI to **augment** (not replace) humans ³³ . Shneiderman proposes frameworks like **governors** (keeping human control), **metaphors** for explainability, and multi-level oversight to ensure AI systems are reliable, safe, and aligned with human values. (Balances technical AI insights with design principles and ethics.)
- **Designing Agentic Technology: AI That Works for People** – Christopher Noessel (2017). A practical book on creating “agentic” UX – where AI agents perform tasks on behalf of users. Noessel cuts through AI hype and offers concrete design patterns for automation, from calendars that schedule themselves to email triage assistants ³⁴ . Emphasizes giving users appropriate control and feedback when delegating tasks to AI.
- **Big Data, Big Design: Why Designers Should Care About AI** – Helen Armstrong (2021). A designer’s primer on machine learning, data, and design. Through case studies and interviews, Armstrong illustrates how **human-centered design** can harness AI and “*big data*” for good ³⁵ . Topics include designing data visualizations, ethical dataset curation, and collaborating with data scientists. It’s a friendly introduction to data/AI concepts for UX practitioners, stressing that designers have a key role in making AI understandable and humane.
- **AI for UX Designers: Using Artificial Intelligence to Supercharge Your Workflow** – Lise Politt (2024). A forthcoming guide (noted in design forums) that helps UX professionals integrate AI into the **design process** ³⁶ . It covers using AI tools for user research (e.g. pattern mining from feedback), generative design (creating variations with AI), and even prototyping with AI-driven content. Also discusses how designers can contribute to AI product teams by understanding data and algorithms – framing AI as both a *tool* in our toolbox and a design *material* in products.
- **The Design of Human-Centric AI** – (Various authors, 2020s). An emerging body of academic work (e.g. Springer’s 2022 HCAI collection) focuses on intelligent agents in real domains like healthcare, education, and finance ³⁷ . These papers propose design principles for AI in the workplace (e.g. physician-facing diagnostic AI) and methods like “*participatory design for AI*”. A common theme: involving users in the AI design loop and addressing trust, control, and context from the start. *(For a research-oriented reader, these conference papers and journals – CHI, IUI, ACM*

Transactions on Interactive Intelligent Systems, etc. – provide the theoretical underpinnings of AI UX.)

- **Design Justice: Community-Led Practices to Build the Worlds We Need** – Sasha Costanza-Chock (2020). Not AI-specific, but highly relevant to AI design. This book advocates inclusive, equitable design, with marginalized communities co-driving the process ³⁸. Its principles (e.g. “Nothing about us without us”) apply to AI systems which, if designed without diverse input, can reinforce bias. A foundational read for understanding fairness, bias, and social context in tech design – critical for anyone creating AI that impacts people’s lives.
- **Unmasking AI: My Mission to Protect What is Human in a World of Machines** – Joy Buolamwini (2023). Part memoir, part manifesto from the MIT researcher who uncovered racial bias in facial recognition. Buolamwini details her fight for algorithmic justice (like founding the Algorithmic Justice League) and exposes how biased AI harms real people ³⁹. For designers, it’s a powerful reminder of the ethical stakes – motivating human-centered approaches (like auditing AI for bias, and designing UIs that **empower users to question or correct AI**). Joy’s storytelling makes the ethics of AI very accessible and compelling.

AI-Specific UX Guidelines & Design Frameworks

- **Google’s People + AI Guidebook (PAI Guidebook)** – A **free, extensive guide** of best practices for human-centered AI product design from Google’s PAIR team. It covers guidelines in areas like *User Needs & ML*: (start with real user problems, not tech for tech’s sake), *Mental Models*: (help users understand what the AI can/can’t do), *Feedback*: (ensure the AI gives useful, intelligible responses), *Explainability*: (offer reasons for AI outputs), and *Errors*: (plan for graceful failure) ⁴⁰. Each guideline comes with examples (e.g. how to communicate uncertainty in a chatbot’s answer). The PAI Guidebook is a cornerstone resource for designing AI UX; it condenses lessons learned across Google’s AI products (from Google Assistant to Photos) into an actionable checklist for designers.
- **Microsoft’s “18 Guidelines for Human-AI Interaction”** – A set of evidence-based guidelines distilled by Microsoft researchers (Amershi *et al.*, CHI 2019). These 18 rules specify how AI systems *should behave* at four stages of interaction: **Initially** (e.g. make clear what the AI can do), **During use** (e.g. sustain the user’s trust by notifying about changes or confidence), **When Wrong** (e.g. support efficient error correction, offer explanations/apologies), and **Over Time** (e.g. learn from user behavior and adapt) ⁴¹. The guidelines were rigorously validated and are freely available ⁴²; they’ve been adopted in design teams to audit AI features. For instance, one guideline is “Time for Familiarization”: an AI should give users time to learn its quirks initially – implemented in products by offering tutorials or low-stakes trials. Following these 18 best practices helps designers create AI interactions that feel more **predictable, controllable, and trustworthy** for users.
- **IBM’s Design for AI Toolkit** – IBM has published a robust set of resources to ensure **ethically aligned, user-focused AI** in their products. This includes *IBM’s AI Ethics Guidelines* (a framework of principles like **transparency, fairness, accountability, and privacy**), as well as a dedicated “Everyday Ethics for AI” handbook for designers ⁴³. They also integrated AI guidance into their **Carbon Design System** (“Carbon for AI”), providing design components and patterns specific to AI features ⁴⁴. For example, Carbon for AI might include standardized UI elements for an AI assistant’s avatar, confidence sliders, or data privacy notices. IBM’s resources emphasize consistency – *every AI in an IBM product should share a unified foundation of ethical and usable design practices* ⁴⁵ ⁴⁶. Designers can reference these when creating chatbots, predictive tools, etc., even outside IBM, as they address common issues (like avoiding bias in training data or conveying AI limitations clearly to users).
- **Generative AI Design Principles** – With the rise of GPT, DALL-E, and other generative models, new design frameworks are forming. For instance, *Justin Weisz (IBM Research)* proposed **6**

Principles for GenAI UX (2023), including: **Engagement** (make the AI proactive but not annoying), **Transparency** (show source data or reasoning behind generative outputs), **Calibration** (help users calibrate trust by showing uncertainty or offering controls), **Feedback** (let users rate or refine AI outputs), etc ⁴⁷. These principles, shared via Medium and talks, guide designers in creating experiences like coding copilots, AI writing assistants, or image generators. Similarly, startups and design agencies have published guideline sets – e.g. the principle “*Reveal the magic, but hide the complexity*” (showing off what the AI did for the user, but not forcing them to see raw model workings unless needed). As generative AI is so new, **design principles are evolving**, but common advice is to **set proper expectations**, give users **agency to correct or fine-tune** AI outputs, and ensure **safety checks** (preventing harmful content).

- **Conversation & Chatbot UX Guidelines** – Since conversational AI is a major domain, there are specialized guidelines from companies like Amazon (Alexa Voice Design Guide), Google (Assistant Conversational UX guidelines), and Nuance. These cover things like: using the **wake word** politely, confirming ambiguous queries, handling no-match/no-input gracefully, and maintaining personality consistently. For text-based chatbots, frameworks like the **Microsoft Bot Framework Guidelines** advise on multi-turn conversation design, quick reply options, and when to hand off to a human. One example guideline: *always disclose that a user is chatting with AI, not a human*, to set context and trust ⁴⁸. Another: *limit the amount of information given in one response* to avoid overloading the user (especially in voice). While each platform has specifics, a unifying theme is “**be helpful, be honest, be clear**” – conversational AI should communicate clearly what it can do, ask for clarification when needed, and use a tone appropriate to the brand/use-case (with empathy in support contexts, for example).

Open-Source Design Toolkits & UI Kits for AI Interfaces

- **Carbon Design System for AI (IBM)** – IBM’s Carbon is an open-source design system, and its “*Carbon for AI*” extension provides UI components tailored to AI interactions ⁴⁴. For example, Carbon includes patterns for **confidence indicators** (so an AI can show uncertainty), **explanation modals** (a standardized way for an AI to explain a recommendation), and **chatbot dialogs** within apps. Designers implementing AI features (like an AI-powered dashboard) can use these components to ensure consistency and best practices out-of-the-box. Carbon’s AI kit is freely available, so even non-IBM teams can leverage elements like **bias warnings**, **model switchers** (if offering multiple AI models), or **data visualization templates** that are designed with AI outputs in mind.
- **AI Agent UI Kit (UIdeck)** – A popular community-contributed UI kit for Figma and Tailwind CSS, geared specifically toward AI products ⁴⁹. It includes pre-built layouts for things like **chat interfaces**, **AI assistant overlays**, **result cards with AI-generated content**, and onboarding screens explaining AI features. Open-source kits like this help teams rapidly prototype the front-end of AI apps – one can grab a “ChatGPT-style” chat window, a voice command button, or a result box that shows an AI answer alongside source links. By starting from these patterns, designers ensure they’re using UI conventions users now expect from AI tools (for instance, the **typing indicator** animation when the AI is “thinking,” or the option to **regenerate** an answer).
- **Conversation Design Toolkits** – There are also free libraries for more specific AI UX needs. E.g., Google released a **Material Design Conversation** kit a few years back, which provided guidance on bubbles, turn-taking visuals, and error prompts for chat UIs. Similarly, the open-source **Bot Framework Web Chat** by Microsoft offers a customizable webchat component (UI) supporting rich cards, quick replies, and speech. Designers can fork these to match their style while inheriting solid interaction patterns. Another example is **Voice UI kits** – Amazon provides an **Alexa Design Guide** with sample utterance flows and situational voice responses (for confirming actions, handling misunderstandings, etc.). Using these toolkits saves time and brings proven usability to AI interfaces.

- **Figma Community Resources** – The Figma community has shared numerous AI-related design files. For instance, templates for **AI onboarding tooltips** (to educate users about new AI features), icon sets for AI (common symbols for “AI suggestion” or “auto-complete”), and **prompt editor** UIs are available. Some designers have open-sourced their explorations of “what would a good settings panel for AI look like?” including toggles for model versions or privacy levels. There are even Figma plugins now (e.g. **DesignerBot**) that can generate UI designs with AI – illustrating how AI is both the subject and the tool in modern design workflows. The key takeaway: a wealth of free resources exists to jumpstart designing AI experiences, and the community is actively iterating on these patterns as the technology evolves.

Courses & Video Lectures on AI UX Design

- **Design for a Better World – IxDF Master Class** – Mentioned earlier, this course with Don Norman (offered via Interaction Design Foundation) delves into “*humanity-centered design*” and includes modules on designing for emerging tech like AI. Students learn how to apply classic UX methods (user research, prototyping, iteration) to complex socio-technical problems, including AI-driven systems, with an emphasis on ethics and sustainability ³¹ ³². It’s a timely course connecting Norman’s design principles to the age of AI.
- **“AI for UX Design” – Designlab (2024)** – A 4-week intensive course teaching UX designers how to harness AI in their workflow and how to design AI features. Topics include using generative AI for creating design assets, AI-assisted user research (e.g. clustering survey responses with ML), and designing interfaces for AI-powered products (chatbots, recommender systems, etc.). The curriculum culminates in a portfolio project where students design an AI-driven product and get mentorship on considerations like transparency and **failure states** ⁵⁰. (Designlab’s offering reflects a broader trend: many UX education platforms are adding AI-focused modules to keep designers up-to-date.)
- **Microsoft Learn – “Designing UX for AI” Video Series** – Microsoft’s online learning portal has a free 18-part video series on AI UX basics ⁵¹. These short lectures cover things like the 18 AI guidelines, how to prototype an AI chatbot experience, and case studies from Azure Cognitive Services. It’s a great self-paced intro for those who prefer video learning. Likewise, Google’s **UX Community YouTube** has recorded talks (from Google I/O and PAIR summits) such as “*How to Design Interfaces for AI-driven Products*”. These often feature designers walking through real examples (e.g. the design of Gmail’s “Smart Compose” feature) and explaining the design decisions and user testing insights behind them.
- **Academic Courses & HCI Workshops** – Universities have begun offering courses explicitly on Human-AI Interaction. For example, Stanford’s CS 376 “**Human-Centered AI**” seminar (and their **HAI** institute talks) bring researchers and students together to discuss cutting-edge papers on explainable AI and AI ethics in UX. MIT’s Media Lab has workshops on “*Designing for Conversational Agents*.” Additionally, conferences like ACM CHI and DIS often run **tutorials** or workshops for practitioners – e.g. “*From UX to AIX: Teaching Designers to Think AI*” (CHI 2020 workshop) – where one can learn the mindset shift required when designing AI (such as focusing on training data, probabilistic outcomes, and long-term learning cycles). Many of these academic lecture videos and materials are available online.
- **Webinars & Community Events** – The UX community on platforms like *UXPA*, *IxDA*, and *UX Collective* frequently host webinars or panel discussions on AI. One example is a UXPA webinar “*Designing the UX for AI: Not Your Typical UX*”, which discussed how designing an AI system differs from normal apps (highlighting issues of **trust, unpredictability, and the need for error recovery**) ⁵². Another is NN/g’s recent virtual conference where a talk titled “*A Research Agenda for Generative AI in UX*” ⁵³ was presented, covering open questions and opportunities for UX professionals in the GenAI space. Attending or watching recordings of these community events

can provide practical tips (often Q&A with experts) and reassurance that everyone is collectively figuring out AI's UX challenges.

Practical Resources (Prompt Design, Explainability, Trust & More)

- **Prompt Engineering Guides & Patterns** – With the advent of large language models (LLMs) and image generators, *prompt design* has become a critical UX element. OpenAI's official guide on prompt engineering (for GPT-4/API) offers best practices – e.g. **provide clear instructions, give examples in the prompt**, and **specify the format you want** ⁵⁴. They enumerate successful prompt **formats** (like bulleting steps for an AI, or framing requests as Q&A) that designers and writers can use when crafting system prompts or user-facing prompts. Beyond official docs, the community has created *Prompt Pattern Catalogs* ⁵⁵ – compilations of reusable prompt templates (for instance, the “XYZ” pattern: “*Explain like I’m a [persona]*” or the “Do and Criticize” pattern where the AI tries a solution then critiques itself). Such catalogs, often on GitHub or papers ⁵⁶, help in designing interfaces that assist users in phrasing their inputs effectively. For example, a creative writing app might implement a **prompt builder UI**: guiding the user to input the setting, character, and tone, which then forms a prompt – these builder interfaces draw on known prompt patterns to improve AI output quality.
- **Explainability & Trust-Calibrating UX** – A major practical challenge is how to present AI decisions or recommendations to users in a transparent way. Design resources suggest **UI elements for explainability**: e.g. “*Why did you get this result?*” links or info icons that expand into an explanation panel. Research shows that simple, *scrutable explanations* (like “*Recommended because you watched X*” on Netflix) can help users understand and trust AI more appropriately ⁵⁷. Another pattern is showing **confidence scores** or levels (e.g. an email spam filter indicating 95% sure of spam) – but the design must be careful: users might misinterpret numbers, so sometimes a qualitative confidence indicator (like a phrase “*I’m somewhat confident in this answer*” or a visual meter) is better. To **calibrate trust**, guidelines say to avoid false certainty; if the AI isn't sure, admit it. For instance, Microsoft's guidelines encourage phrasing uncertain outputs with **hedging** (“I think...”). Also, allowing users to give feedback on AI outputs (a thumbs up/down, or edit the result) serves two purposes: it gives users agency and it gathers data to improve the AI. In practice, a well-known example is Google's *Smart Reply* – they found users trust it more when it occasionally explains it's using common phrases or when it offers multiple options (so the user doesn't think the single suggestion is always “*the correct*” one). **Summary**: Many practical UI techniques (tooltips, colored confidence bars, explanation text, feedback buttons) are available to make AI more **transparent and trustworthy**, and these are documented in both research and industry playbooks ⁴⁸ ⁵⁸.
- **“Prompt Augmentation” Design Patterns** – As identified by Jakob Nielsen (2023) and others, there is an “*articulation barrier*” for users in formulating perfect prompts to an AI ⁵⁹. To help bridge the gap between user intent and AI output, designers are creating features we can call **prompt augmentation**. Six emerging UX patterns in this vein are: **Style Galleries** – offering a gallery of styles or examples that users can pick to refine AI output (common in image generators: e.g. choosing “*Van Gogh style*” or “*1960s photo*” to append to your prompt) ⁶⁰ ⁶¹; **Prompt Rewrite** – the system suggests alternative phrasings of a user's request (like “*Try asking in a more specific way*” with a ready-made tweak); **Targeted Prompts** – UI shortcuts for common modifications (e.g. buttons: “make it funnier”, “simplify the language” in an AI writing assistant); **Related Prompts** – suggestions of what to ask next (to guide a user who isn't sure how to continue the dialog); **Prompt Builders** – form-like interfaces that break a complex prompt into fields (e.g. a DALL-E form: *Subject? Style? Lighting?* which then combines into the final prompt); and **Parameter Tuning UIs** – sliders or toggles for advanced parameters (like creativity level, or excluding certain content). These patterns, documented in Nielsen's article and other UX blogs, have already appeared in many AI tools. For example, OpenAI's ChatGPT UI added a

“Regenerate response” button – which is essentially a quick *prompt rewrite* trigger. Adobe’s generative image UI provides **style filters** (a kind of style gallery). By including prompt augmentation features, designers can significantly improve the average user’s outcome with AI, as they no longer have to craft a perfect sentence from scratch – the interface guides them in expressing their intent more effectively ⁶² ⁶³ .

- **Error Handling & User Guidance** – Practical design tips for AI often focus on edge cases: what if the AI **fails** or produces wrong/harmful output? Unlike traditional software, AI can fail unpredictably, so UX must handle this gracefully. Common patterns: **detecting confusion** – if the AI’s confidence is very low or it appears to “hallucinate,” apologize and either refrain from answering or flag the content. For instance, Bing Chat will sometimes say *“I’m sorry, I don’t have enough knowledge to help with that.”* Guidelines suggest providing alternative next steps in such cases (perhaps *“Would you like to search the web for this?”* or offer to connect to a human agent) ⁴⁸ . Another aspect is **user education**: since AI is new to many users, interfaces often include *tutorials, tooltips*, or even subtle cues to teach users how to interact. For example, a prompt input box might have placeholder text like *“Ask me to draft an email, or type @ to see examples”*, indicating capabilities. Some apps have a “tour” that highlights: *“This is an AI feature. It learns from your feedback – use the thumbs up/down to refine its suggestions.”* Practical resources (like OpenAI’s documentation and community forums) provide text snippets and UX copy recommendations for these scenarios. Also, error states should avoid blaming the user (“I didn’t understand *your* question”); instead they should take on the blame or offer help (“Sorry, I’m not sure how to answer that. Maybe try rephrasing or adding detail.”). Overall, designing robust error handling in AI UX is crucial to keep user trust – it’s better to **fail gracefully** and help the user move forward than to give a confusing or confident-but-wrong answer with no support.

Communities, Conferences & Influencers in AI UX

- **Academic & Industry Conferences** – The intersection of AI and UX has become a hot topic at gatherings like **ACM CHI (Conference on Human Factors in Computing)** and **IUI (Intelligent User Interfaces)**. These conferences have entire tracks on *human-AI interaction*, where researchers share findings on UX techniques for AI. For example, CHI 2023 featured papers on how UI design affects users’ trust in AI, and IUI often hosts demos of novel interfaces (like an AR assistant that explains machine decisions in real-time). There are also newer conferences specifically on *Human-Centered AI* (e.g. Stanford HAI’s annual symposium) and *Fairness/Accountability in AI (ACM FAccT)* which, while more academic, influence the design guidelines tech companies adopt. Designers interested in deep dives can follow these venues; many papers are open-access or discussed in summary articles.
- **Online Communities & Forums** – The design community has rallied to discuss AI on platforms like **Medium (UX Collective)**, **Reddit**, and **LinkedIn groups**. *UX Collective* has an “AI in UX” series with articles such as *“Don’t build trust with AI, calibrate it”* ⁶⁴ and *“Riding the wave: from human-centered design to human-centered AI”* ⁶⁵ , where practitioners share lessons from real projects. On Reddit, the r/UXDesign sub often has threads on AI (e.g. debating the UX of chatbots like Character.AI or user reactions to Gmail’s Smart Reply). Slack and Discord communities have also formed – for instance, **“AI x Design”** is a Slack community where designers brainstorm solutions to AI UX problems, share interface screenshots, and even critique each other’s prompts and prototypes. These peer communities are great for staying current, as someone will inevitably post the latest “ChatGPT just added this new feature, how might we leverage that?” or “Check out this failure – DALL-E gave a bizarre result, how should the UI handle it?”
- **Leading Voices & Influencers** – A few individuals have become go-to sources for AI+UX insights. **Ben Shneiderman**, mentioned above, is both an academic giant and an advocate for *“Human-Centered AI”* – his talks and articles often outline high-level principles (he’s famously said *“the old computing was ‘garbage in, garbage out’; the new is ‘algorithms in, algorithmic bias out’”*). Don

Norman himself, ever relevant, has started discussing AI's impact – in 2023 he argued designers must extend their scope to global/humanity impacts of AI ²⁷, and cautioned against blindly adopting AI without redesigning systems around it. Saleema Amershi (Microsoft) is a prominent UX researcher who led the 18 Guidelines work; she speaks often on designing developer tools for AI and how to implement the guidelines in practice. Ruth Kikin-Gil (Microsoft Design) writes and presents on “Responsible AI by Design,” offering a practitioner’s perspective on weaving ethical checkpoints into the product design cycle ⁶⁶ ⁶⁷. Josh Lovejoy, formerly at Google’s PAIR, is known for championing inclusive design in ML (for example, designing Pixel phone’s camera to work equitably across skin tones – he’s spoken on that process as a model for AI fairness in UX). On the more popular side, Jakob Nielsen (of NN/g), one of UX’s founders, has turned his attention to AI UX lately – his March 2023 article on Prompt Augmentation ⁵⁹ gained wide notice, and when Nielsen highlights AI, many in the UX industry take note due to his credibility from the usability era. Joy Buolamwini (technologist/activist mentioned prior) isn’t a UX designer per se, but her influence on awareness of AI bias has made algorithmic fairness a design consideration; she keynoted at SXSW and Grace Hopper Conference, inspiring designers to consider “Who might this AI harm, and how could we design safeguards?”*.

- **Meetups, Workshops, and Hackathons** – Many cities now have meetups blending AI and design. For example, in San Francisco there’s “**AI UX Meetup**” which hosts monthly talks – one month a designer from OpenAI might discuss ChatGPT’s UI challenges, another month a researcher might run a tutorial on prototyping voice assistants. Globally, events like **Service Design Global Conference 2023** had sessions on AI in service blueprints, and **ProductTank** meetups have featured “Designing AI Products” talks. **Hackathons** too have embraced AI UX: e.g. hackathons where teams must design a solution using an AI API *and* pay special attention to user experience (often with a prize for best usability). These grass-roots events are great for hands-on learning and networking – one might meet a “prompt engineer” to collaborate with, or a user researcher exploring how to test AI features.
- **Cross-disciplinary Collaboration** – Finally, the AI design community often intersects with data science and AI research communities. Initiatives like “**ML+Design**” workshops bring together UX designers and ML engineers to foster mutual understanding. Conferences such as **NeurIPS (a top AI research conference)** have started including *Improving Human-AI Interaction* as a topic in their workshops, drawing UX people into traditionally technical spaces. Online, the **Partnership on AI** (an organization) runs working groups that include designers, AI scientists, policy folks, etc., to create reports on things like **AI transparency** and **explainability guidelines**. Being aware of these interdisciplinary efforts is useful – they often yield free toolkits or reports. For instance, Partnership on AI released **UX guidelines for AI explainability** in 2021 (a report compiled with input from companies and nonprofits). By tapping such resources, designers can stay ahead of regulatory expectations (like the EU AI Act’s transparency requirements) and ensure their work aligns with broader standards emerging around AI.

In summary, the landscape of AI design and UX is rich and rapidly evolving. From **foundational theory** (Don Norman’s human-centered ethos and HCI research) to **current best practices** (AI interaction guidelines, ethical design frameworks) to **practical tools** (UI kits, prompt libraries) to **communities of practice** (where designers share and learn together), there is a wealth of resources. Designers facing **GenAI and AI-driven** interface challenges today can draw on these to create products that are not only innovative, but also **usable, trustworthy, and aligned with human needs** – carrying forward the legacy of human-centered design into the AI age.

Sources: Don Norman’s JND.org (books & essays) ⁶⁸ ⁶⁹; Nielsen Norman Group (book summaries & articles) ³ ⁴ ¹⁷; McKinsey interview with D. Norman ²⁰; UX Magazine interview ²⁴; Google PAIR Guidebook ⁴⁰; Microsoft Research guidelines ⁴¹; IBM Design for AI ⁴³; Jakob Nielsen (UX Tigers) on AI patterns ⁵⁹; Salesforce UX blog (AI books list) ³⁵ ³⁹; and assorted research/industry publications as cited above.

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- 1 **Design for a Better World**
<https://mitpress.mit.edu/9780262548304/design-for-a-better-world/>
 - 2 20 23 27 **Author Talks: Don Norman on humanity centered design | McKinsey**
<https://www.mckinsey.com/featured-insights/mckinsey-on-books/author-talks-don-norman-designs-a-better-world>
 - 3 29 53 **The Design of Everyday Things: Revised and Expanded Edition - NN/G**
<https://www.nngroup.com/books/design-everyday-things-revised/>
 - 4 30 **Emotional Design | Book by Don Norman - NN/G**
<https://www.nngroup.com/books/emotional-design/>
 - 5 **Living with Complexity – Don Norman's JND.org**
<https://jnd.org/books/living-with-complexity/>
 - 6 **The Design of Future Things – Don Norman's JND.org**
<https://jnd.org/books/the-design-of-future-things/>
 - 7 **The Invisible Computer: Book by Don Norman - NN/G**
<https://www.nngroup.com/books/invisible-computer/>
 - 8 **Things That Make us Smart (Book by Don Norman) - NN/G**
<https://www.nngroup.com/books/things-that-make-us-smart/>
 - 9 **Turn Signals Are the Facial Expressions of Automobiles | by Don Norman - NN/G**
<https://www.nngroup.com/books/turn-signals-are-the-facial-expressions-of-auto/>
 - 10 **Don Norman - Wikipedia**
https://en.wikipedia.org/wiki/Don_Norman
 - 11 12 **User Centered System Design: New Perspectives on Human-computer Interaction**
https://www.routledge.com/User-Centered-System-Design-New-Perspectives-on-Human-computer-Interaction/Norman-Draper/p/book/9780808598728?srsltid=AfmBOoop4RoiOWgoY2fxaHcSu_DRLi_W-m7AlrEgaLW2UlyYno8WVfhV
 - 13 14 **Norman, Don | Faculty | Northwestern Engineering**
<http://www.mccormick.northwestern.edu/research-faculty/directory/profiles/norman-don.html>
 - 15 **Unix Philosophy | Encyclopedia MDPI**
<https://encyclopedia.pub/entry/28459>
 - 16 **HCD harmful? A Clarification – Don Norman's JND.org**
<https://jnd.org/hcd-harmful-a-clarification/>
 - 17 18 **Design Thinking: A Useful Myth - NN/G**
<https://www.nngroup.com/articles/design-thinking-useful-myth/>
 - 19 **Don Norman: 3 ways good design makes you happy | TED Talk**
https://www.ted.com/talks/don_norman_3_ways_good_design_makes_you_happy
 - 21 **Oral History Interview with Don Norman | ACM Other Books**
<https://dl.acm.org/doi/full/10.1145/3505178>
 - 22 **Interviews – Don Norman's JND.org**
<https://jnd.org/category/Interviews/>
 - 24 25 26 **Solving the Right Problem and Finding Your Own Solution: An Interview with Don Norman - UX Magazine**
<https://uxmag.com/articles/solving-the-right-problem-and-finding-your-own-solution-an-interview-with-don-norman>

28 68 Books – Don Norman's JND.org

<https://jnd.org/books/>

31 32 Design for a Better World with Don Norman | IxDF

https://www.interaction-design.org/courses/design-for-a-better-world-with-don-norman-course?srsId=AfmBOopmcs2n2hrxZoNGclShq_1k_QbX8_edqWE4lEtYz1o-leEkkY5y

33 Human-centered AI - Ben Shneiderman - Google Books

https://books.google.com/books/about/Human_centered_AI.html?id=YS9VEAAQBAJ

34 Designing Agentive Technology: AI That Works for People

<https://www.amazon.com/Designing-Agentive-Technology-Works-People/dp/1933820632>

35 36 38 39 10 Books for UX Designers That Boost Your AI Mindset

<https://www.salesforce.com/blog/10-best-ai-design-books/>

37 The Design of Human-Centered Artificial Intelligence ... - SpringerLink

<https://link.springer.com/book/10.1007/978-3-031-83512-4>

40 People + AI Guidebook - Home

<https://pair.withgoogle.com/guidebook/>

41 42 66 67 Guidelines for Human-AI Interaction - Microsoft Research

<https://www.microsoft.com/en-us/research/project/guidelines-for-human-ai-interaction/>

43 45 46 IBM Design for AI

<https://www.ibm.com/design/ai/>

44 Carbon for AI

<https://carbondesignsystem.com/guidelines/carbon-for-ai/>

47 Design Principles for Generative AI Applications | by Justin Weisz

<https://medium.com/design-ibm/design-principles-for-generative-ai-applications-791d00529d6f>

48 5 UX Rules for Building Trust in Enterprise AI - EMERGE

<https://www.emergeagency.com/insights/detail/ai-machine-learning-best-practices-enterprise-applications/>

49 AI Agent: AI UI Kit Library for Tailwind CSS and Figma - UIdeck

<https://uideck.com/templates/ai-agent>

50 AI for UX Design Course | Designlab

<https://designlab.com/advanced/ai-for-ux-design-and-product-design>

51 Designing UX for AI Applications (Part 12 of 18) - Microsoft Learn

<https://learn.microsoft.com/en-us/shows/generative-ai-for-beginners/designing-ux-for-ai-applications-generative-ai-for-beginners>

52 Designing the UX for AI: Not Your Typical UX Design

https://uxpa.org/aiovg_videos/designing-the-ux-for-ai-not-your-typical-ux-design/

54 Best practices for prompt engineering with the OpenAI API

<https://help.openai.com/en/articles/6654000-best-practices-for-prompt-engineering-with-the-openai-api>

55 [PDF] A Prompt Pattern Catalog to Enhance Prompt Engineering with ...

<https://www.dre.vanderbilt.edu/~schmidt/PDF/prompt-patterns.pdf>

56 Prompt patterns for LLMs that help you design better software

<https://chuniversity.nl/papers/prompt-patterns-for-software-design>

57 Explainable recommendation: when design meets trust calibration

<https://pmc.ncbi.nlm.nih.gov/articles/PMC8327305/>

58 Assistive to Agentic AI: Risks, Responsibilities, and the Road Ahead

<https://news.sap.com/2025/06/assistive-agentic-ai-risks-responsibilities-road-ahead/>

59 60 61 62 63 Prompt Augmentation: UX Design Patterns for Better AI Prompting

<https://www.uxtigers.com/post/prompt-augmentation>

64 Don't build trust with AI, calibrate it | by Irina Nik | UX Collective

<https://uxdesign.cc/dont-build-trust-with-ai-calibrate-it-2889a5740e16>

65 Riding the wave: from human-centered design to human-centered AI

<https://uxdesign.cc/riding-the-wave-from-human-centered-design-to-human-centered-ai-5ab0f3b321c9>

69 The Design of Everyday Things (first edition) – Don Norman's JND.org

<https://jnd.org/books/the-design-of-everyday-things-1st-ed/>